

CS-302: Artificial Intelligence

LECTURE NO 11:

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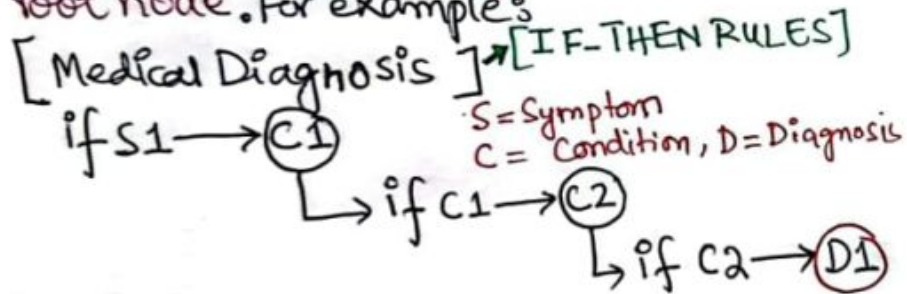
Organization of Lecture

1. Comparison between Forward and Backward Chaining/Reasoning
2. Significance of Inference Engine in Expert Systems
3. Forward chaining and backward chaining Strategies
4. Representing Class Membership and Class Inclusion in predicate logic
5. Comparison of Proposition and predicate Calculus
6. Min Max Algorithm

1. Comparison between Forward and Backward Chaining/Reasoning

Forward Reasoning: It moves forward from Start to goal state. Also called as Data Driven Reasoning.

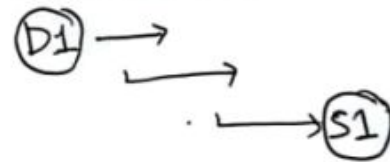
- Search Tree has Initial Configuration(s) at root of tree
- Next level of tree is generated by finding all the rules whose left side matches the root node. For example:



- ① Initially system is provided with one or more constraint. In above case it is 'S1'.
 - ② Rules are searched for each constraint.
 - ③ Satisfying rules (R.H.S) becomes (L.H.S).
i.e., R.H.S (then part), L.H.S (if part)
- This loop will stop until there are no more rules.

Backward Reasoning: It moves backward from goal to initial state. Also called as Goal Driven Reasoning

- Search Tree has Goal Configuration(s) at the root of the tree.
- Next Level of Tree is generated by finding all the rules whose right side matches the root node. For example:



- ① First, Goal state and rules are selected, where the Goal state resides in THEN (R.H.S) part. (L.H.S)
- ② From if part of selected rules subgoals are made.

2. Significance of Inference Engine in Expert Systems

→ Inference Engine acquires and manipulates knowledge from the Knowledge Base to arrive at particular solution.

Rule Based Expert Systems

- i) Applies Rules repeatedly to facts.
- ii) Add new knowledge to Knowledge Base.
- iii) Resolve Rule Conflicts

Rules generated from earlier application

3. Forward chaining and backward chaining

Forward chaining:

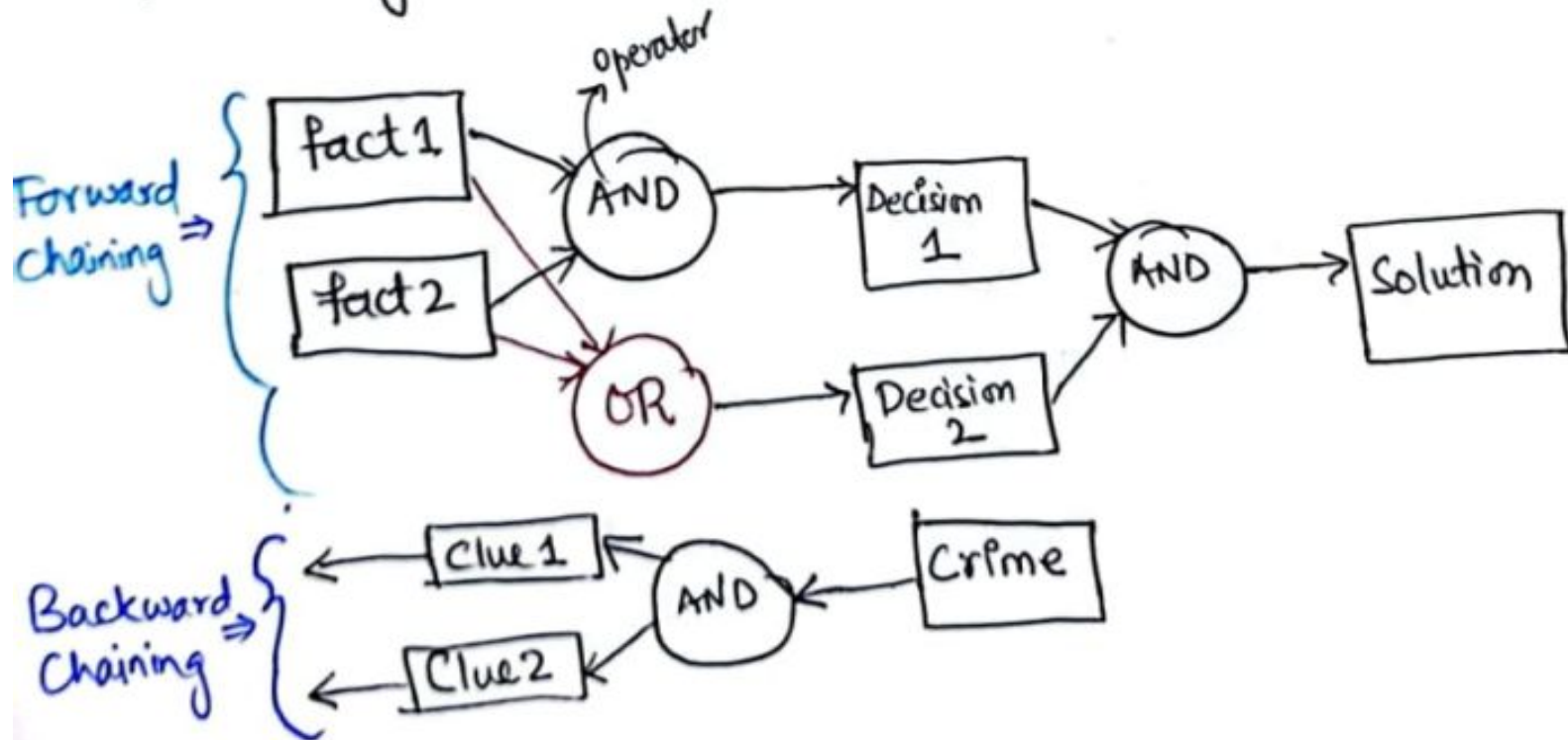
↳ What can happen next?

↳ follows chain of conditions & deduce the outcome.

Backward chaining:

"Why this happened?"

↳ Finding out cause/reason.



4. Representing Class Membership and Class Inclusion in predicate logic (1/2)

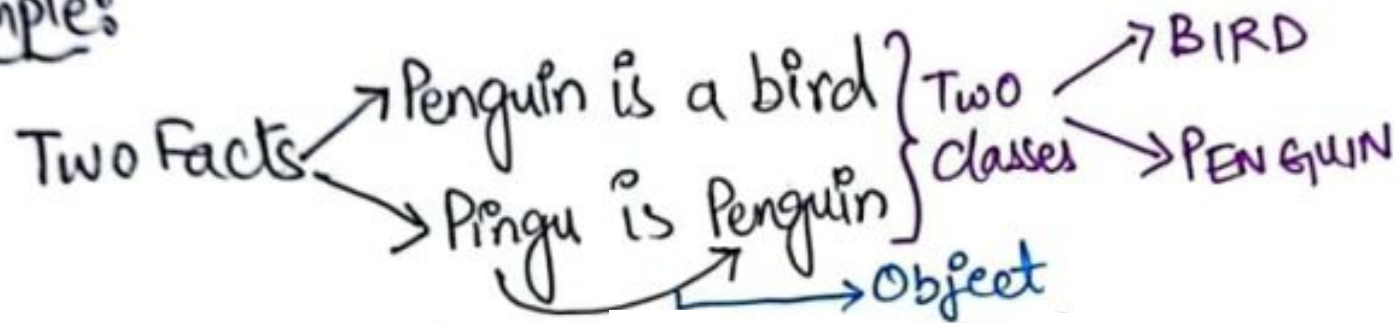
④ Representing class Membership and class Inclusion in predicate logic:

Two predicates Isa and Instance are used.

Represents Relationship of class inclusion

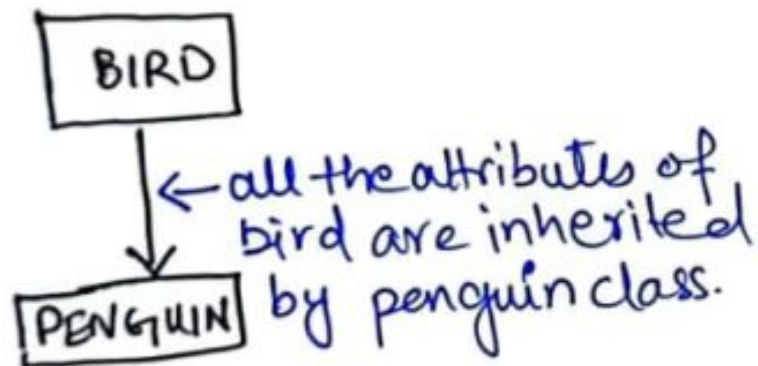
Represents class membership relationship.

Examples



4. Representing Class Membership and Class Inclusion in predicate logic (2/2)

→ We can say that objects of penguin class will inherit all the attributes/properties of the class BIRD.



→ Penguin class is nothing but a inclusion of bird class.
i.e.,

Class(Penguin, Bird) { is a }

→ Pingu is an ^{instance} object of class penguin. i.e.,
Instance(Pingu, Penguin) { Instance }

5. Comparison of Proposition and predicate

PREDICATE CALCULUS

- Predicates are used that involves constants, variables, relation, function.

- It can represent individual entities.

* Tall(Ali).

- FOL can represent generalization, specialization or patterns.

* No-of-Sides (Square, 4)

PROPOSITIONAL CALCULUS

- It uses proposition in which complete sentence is denoted by symbol.

* (It is rainy day) denoted by $\rightarrow X$

- It can't represent individual entities.

* Ali is tall. (We can't individually define Ali, we have to define a complete sentence).

- It can't express generalization, specialization or patterns.

* Square has 4 sides

6. Min Max Algorithm (1/5)

- It is a specialized search algorithm that returns optimal sequence of moves for a player in Zero Sum Game.
- Recursive/Back tracking algorithm which is used in decision making and Game theory.
- Uses recursion to search through Game Tree.
- Algorithm computes min max decision for current state.
- Two Players
 - MAX [Always selects maximum value]
 - MIN [Always selects minimum value]

6. Min Max Algorithm (2/5)

→ Depth-First Search Algorithm is used for exploration of Complete Game Tree.

↳ CHESS, checkers, Tic-Tac-Toe-----

$$\begin{bmatrix} \max = -\infty \\ \min = \infty \end{bmatrix} \begin{matrix} \text{Worst Values} \\ \text{[Initial]} \end{matrix}$$

Properties

① Definitely find solution (if exists)

↳ COMPLETE

② optimal

③ Time Complexity = $O(b^m)$

↳ maximum depth of the tree

↳ branching factor of Game Tree

④ Space Complexity = $O(b^m)$

6. Min Max Algorithm (3/5)

Limitations:

↳ slow for complex games such as Chess

↳ 35 choices/moves

$d = 100$ (for both players)

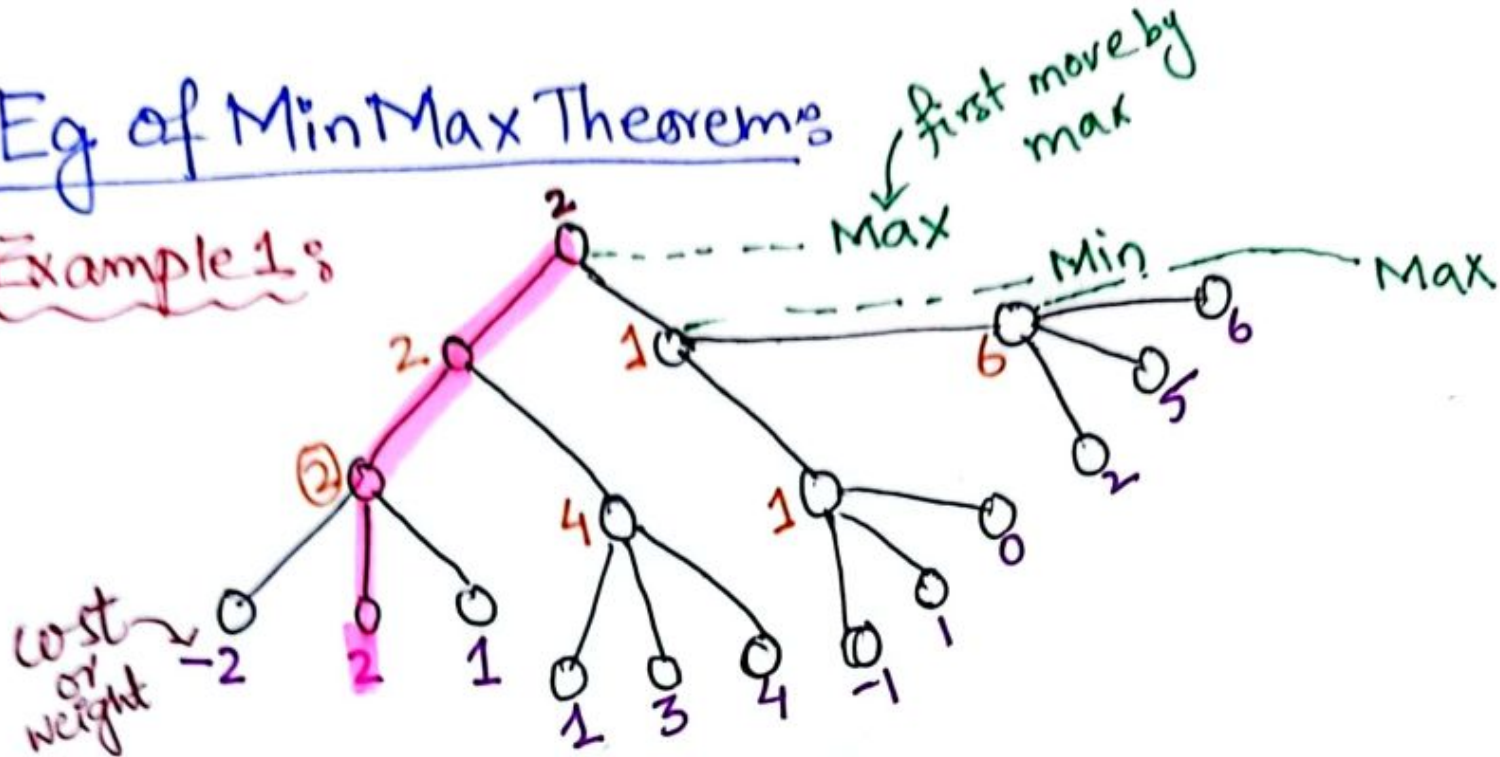
$35^{100} \rightarrow \text{BIG} \leftarrow \text{Time complexity}$

↳ For dealing with this limitation we use/apply Alpha-beta pruning.

6. Min Max Algorithm (4/5)

Eg of MinMax Theorem

Example 1:



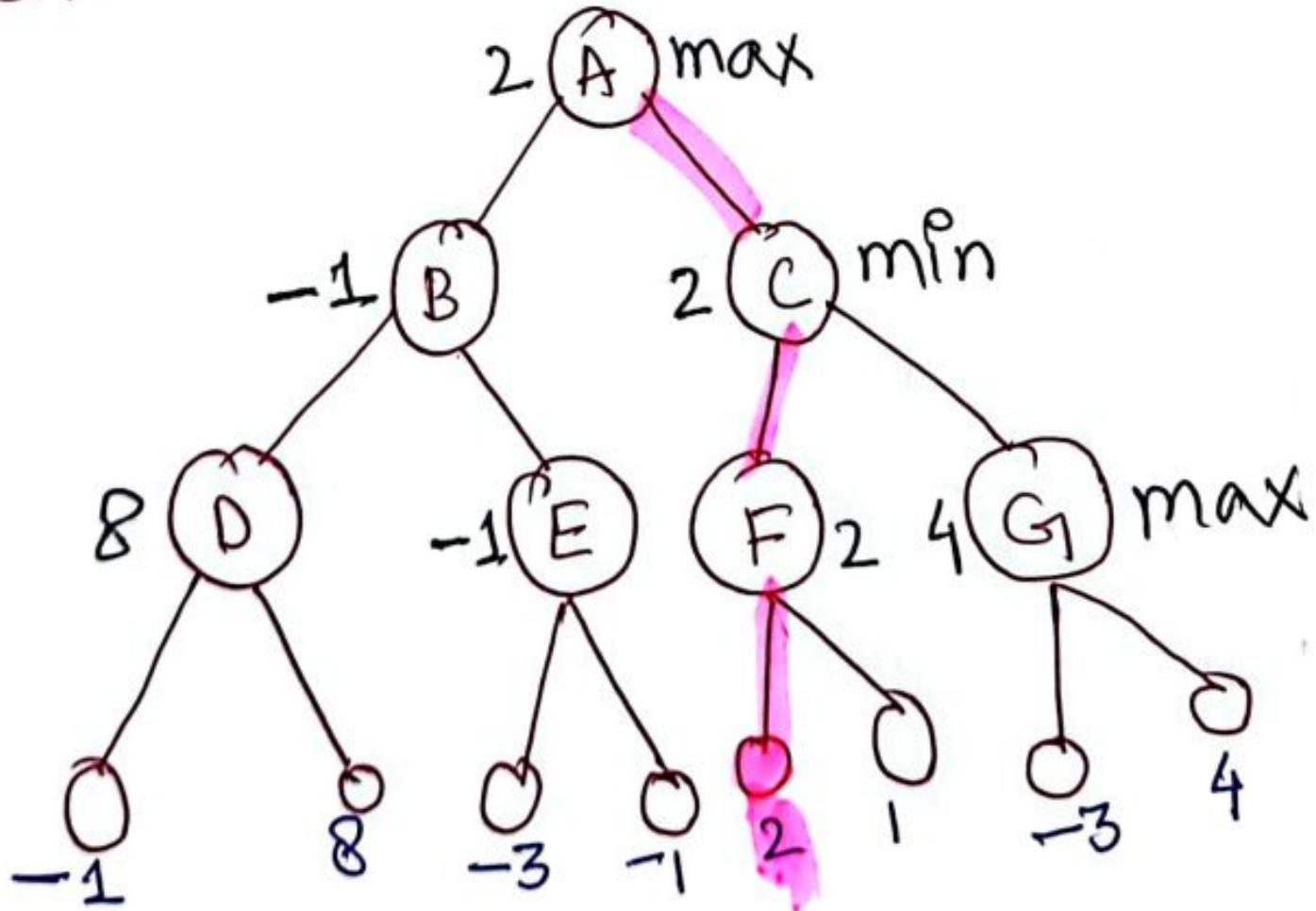
Initial values $\begin{bmatrix} \max = -\infty \\ \min = \infty \end{bmatrix}$

 $(-2, -\infty)$
$$\hookrightarrow (-2, 2)$$
$$L \rightarrow (2, 1)$$

1) $\rightarrow$ 2)

6. Min Max Algorithm (5/5)

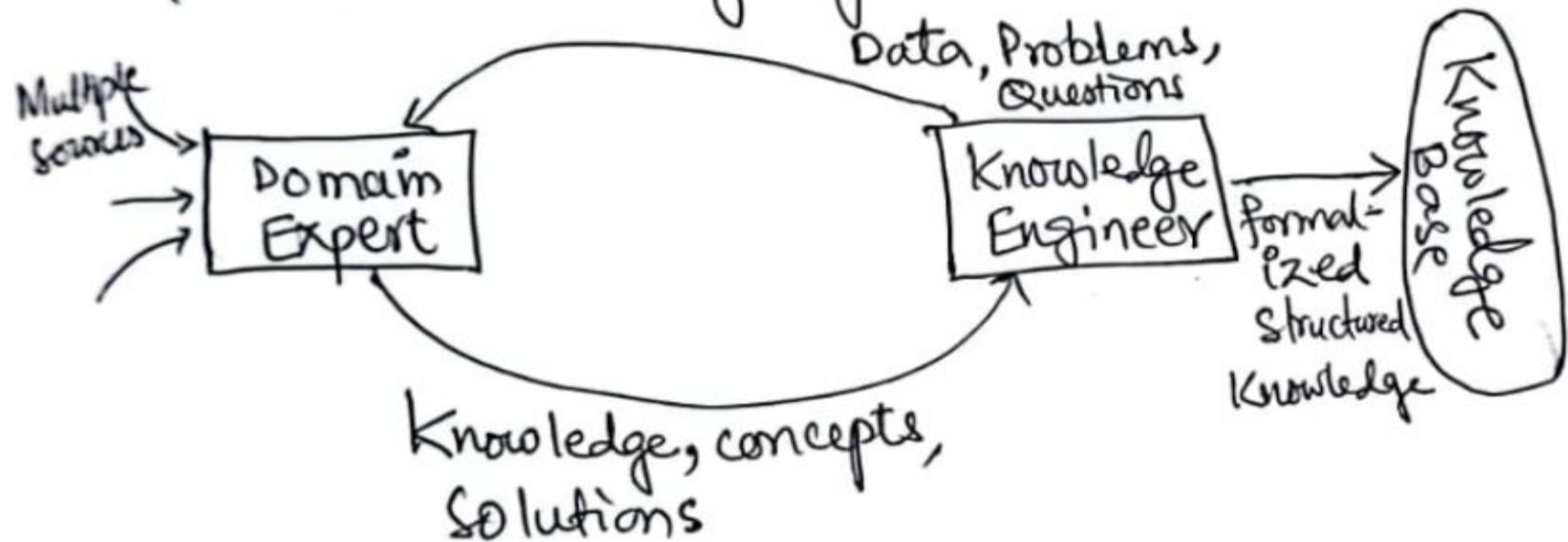
Example 2:



7. Knowledge Acquisition in A.I (1/4)

→ Knowledge Acquisition (KA) is the process of extracting, structuring and organizing knowledge from one or more sources.

→ Knowledge is a collection of specialized facts, procedures and judgement rules.



7. Knowledge Acquisition in A.I (2/4)

Tasks in K A (CIAD)

- i) Collecting: knowledge from expert.
- ii) Interpret: Review & identifying key/main parts
- iii) Analyze: Forming theories, strategies to solve the problem.
- iv) Design: Forming better understanding.

7. Knowledge Acquisition in A.I (3/4)

KA Techniques:

- i) Introspection: Expert acts as both expert & knowledge engineer.
- ii) Observation: Expert closely observes work, i.e., video recording, Analysis.
- iii) Induction: Process of converting set of examples into rules.

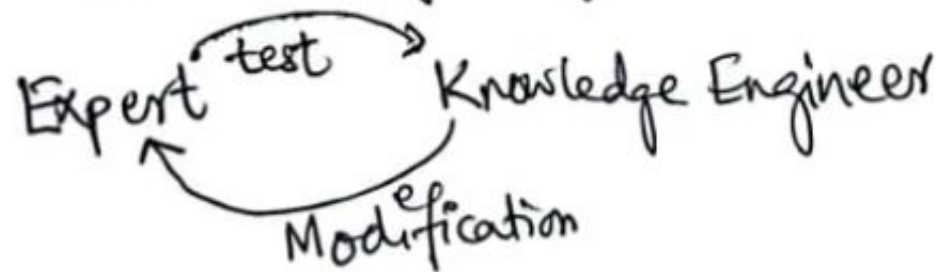
7. Knowledge Acquisition in A.I (4/4)

iv) Protocol Analysis: (Psychology)

↳ Expert is asked to perform a task and verbalize the thought process.

v) Prototyping: Extension of interviewing.

Expert + Knowledge Engineer $\Rightarrow$ Build the System



vi) Interviewing: used in early stages of acquisition. Experts verify the knowledge of an engineer.

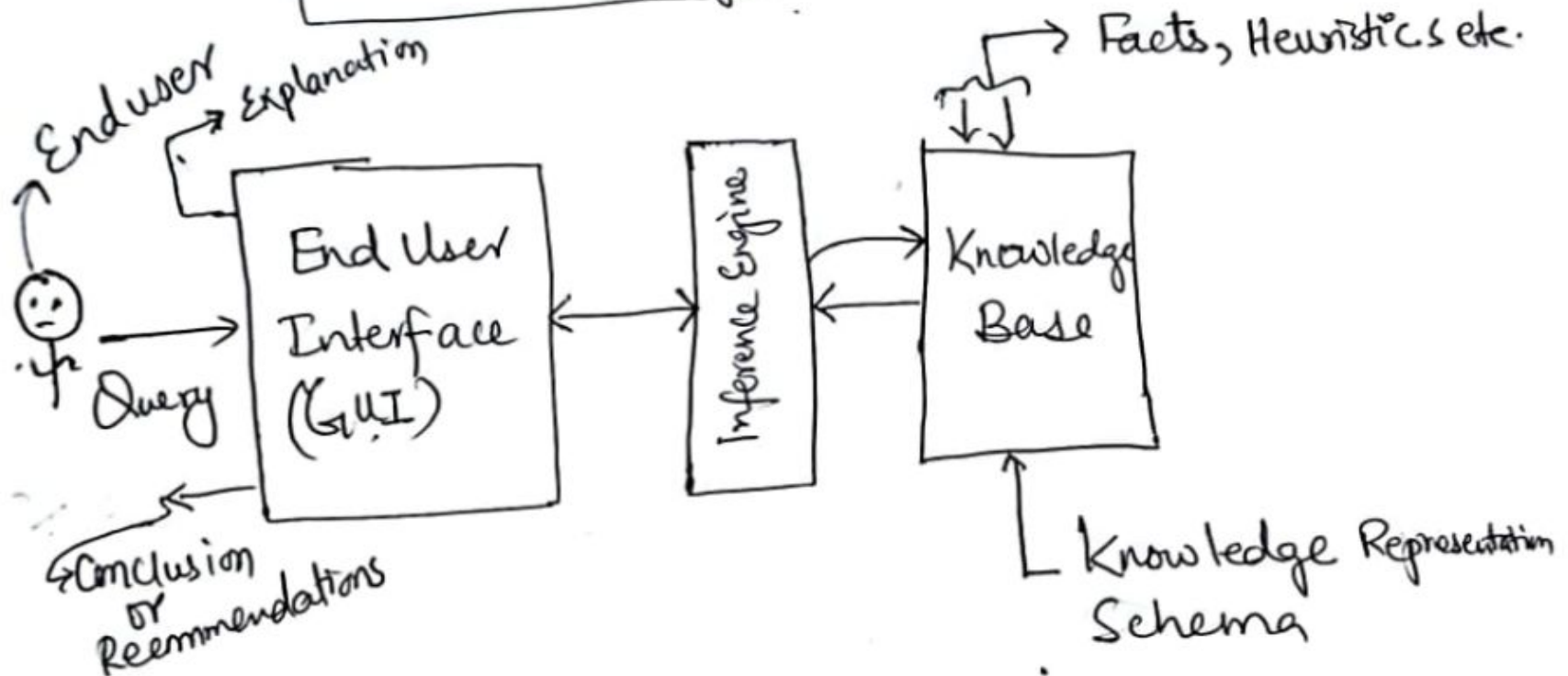
8. Architecture of Knowledge Based System(KBS)

→ KBS is a system that draws upon knowledge of Human Experts in Knowledge Base to solve problems that normally require human expertise.

→ KBS is more general than Expert System.

↳ $\text{KBS} = \text{Knowledge base} + \text{Inference Engine}$

Domain specific knowledge



8. Architecture of Knowledge Based System(KBS)

Components of KBS

- ① Knowledge Base: Contains system organized collection of facts about a system domain.
- ② Inference Engine: It tries to derive answers from Knowledge Base [BRAIN]
- ③ User Interface: Enables the user to communicate with KBS.

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Textbooks

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